

IOT SECURITY

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CONTENTS

CONTEXT

- IOT definition;
- IOT trends,
- IOT innovation and integration;
- Cyber security risks
- IOT contribution to cyber security risk;

RISK

- Main risk overview;
- Security as an afterthought;
- Embedded vulnerabilities;
- Embedded backdoors;
- Unsupported devices;
- Unpatchable devices;

IMPACT

- Main impacts overview;
- IOT used as a bot;
- IOT used to access network;
- IOT used to spy/attack you;
- IOT physical impact;
- IOT Self destruct button;

SECURITY

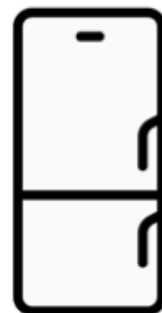
- IOT security overview;
- Traffic analysis;
- Code analysis;
- Firmware analysis;
- Online search;
- Follow best practise;

FUTURE

- Future of IOT

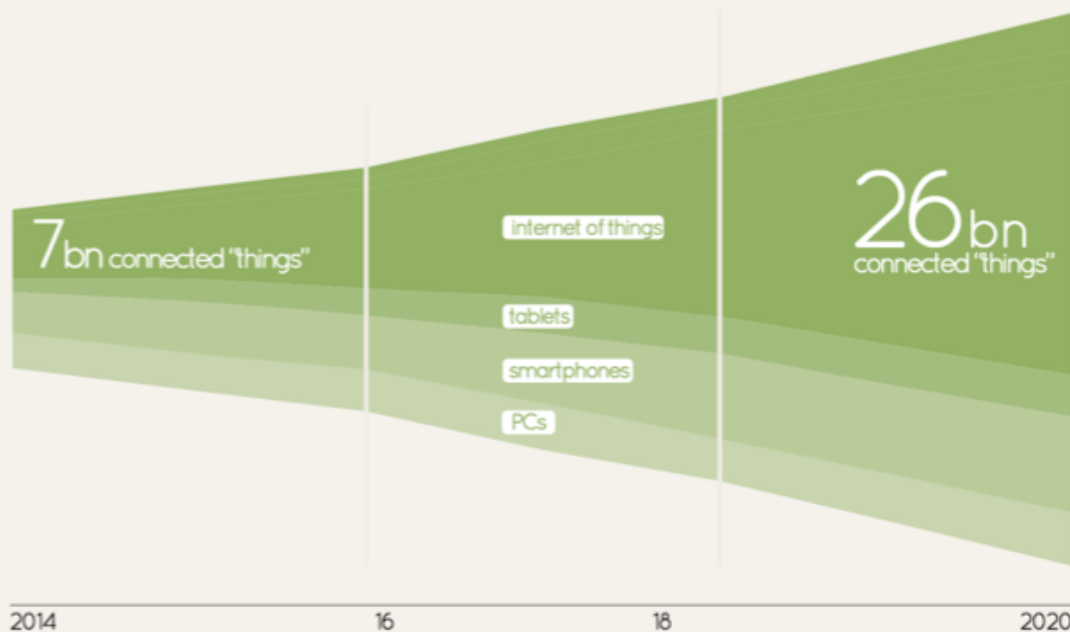
IOT DEFINITION

THE **I**NTERNET **O**F **T**HINGS
ANY PHYSICAL OBJECT THAT CAN BE
CONNECTED TO THE INTERNET



IOT TRENDS

CONNECTED DEVICES



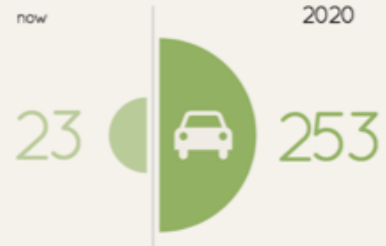
sources: Gartner

IoT MARKET VALUE



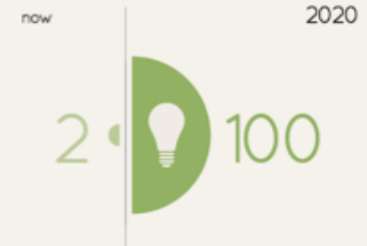
sources: Gartner, Forbes, Cisco, average

CONNECTED CARS millions



source: Forbes

CONNECTED LIGHTS millions



source: On World via Forbes

CONNECTED INDUSTRIES



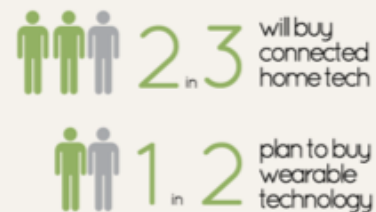
source: Deutsche Bank, Cisco

CONNECTED BIZ



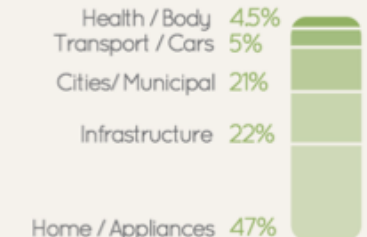
source: SAP

CONNECTED PEOPLE



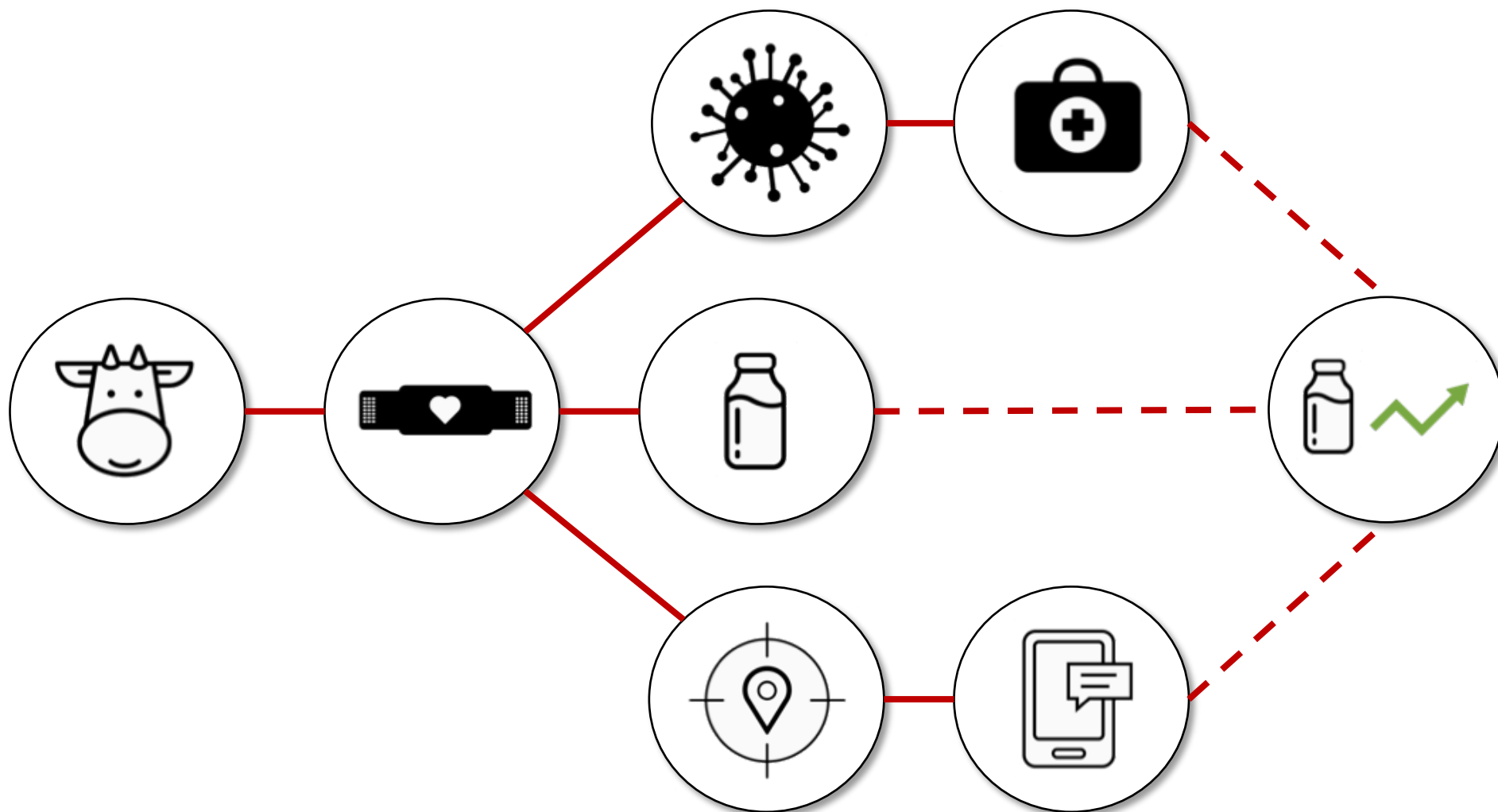
source: Forbes

CONNECTED TECH



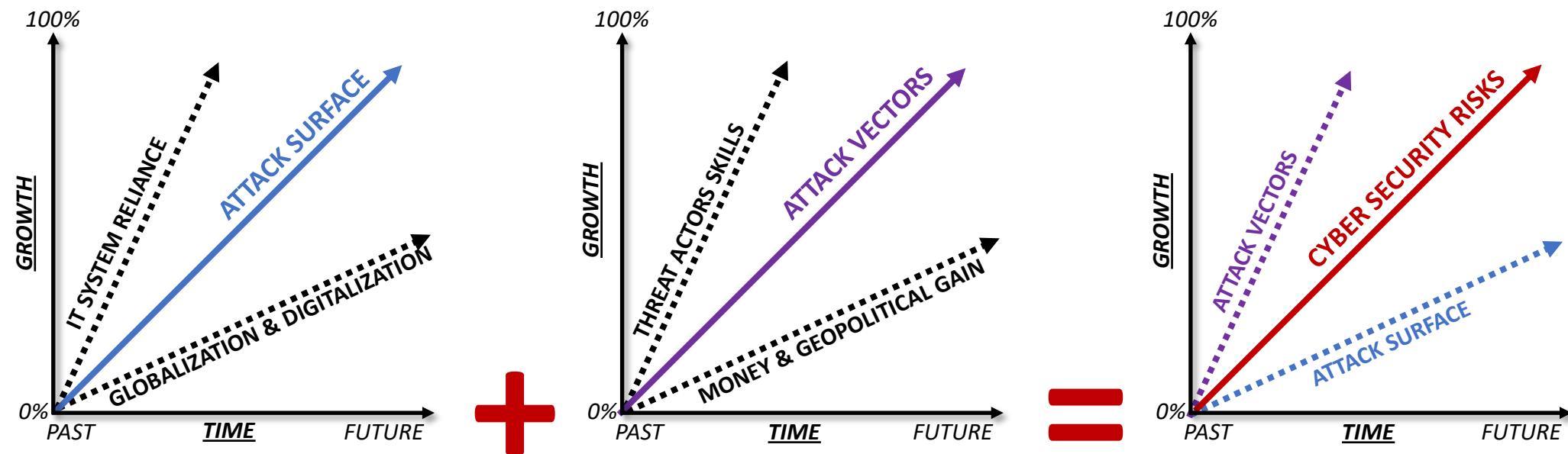
source: Deutsche Bank, Cisco

IOT INNOVATION AND INTEGRATION

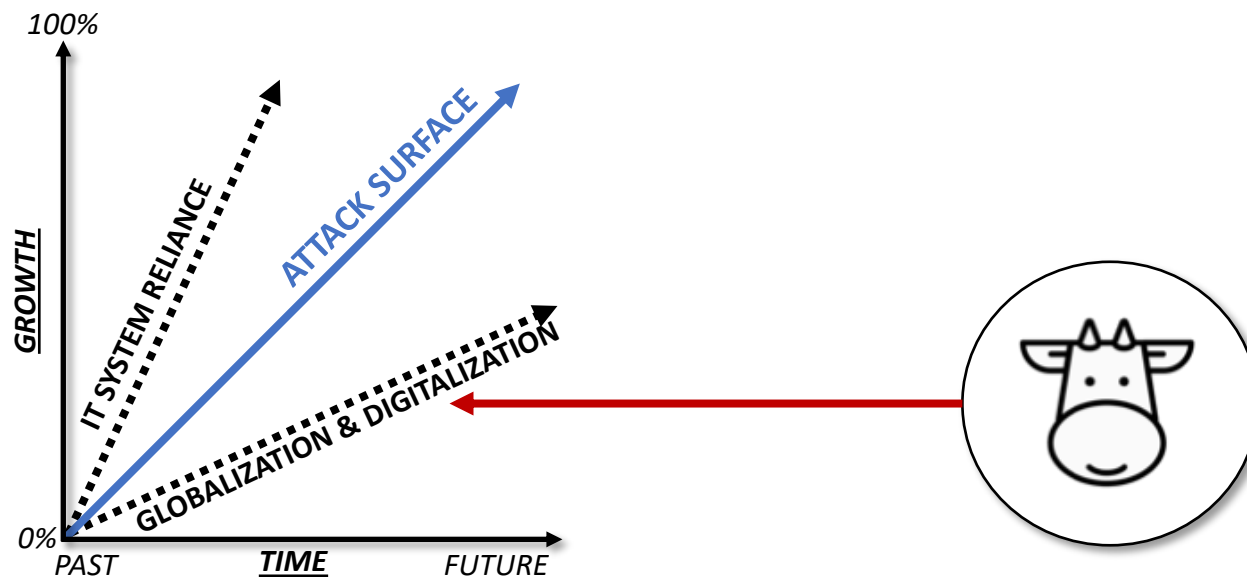


CYBER SECURITY **RISKS' PROBABILITY** AND **IMPACT** ARE **INCREASING**.

THEIR ABILITY TO **DISRUPT** COMPANIES **BUSINESS OPERATION** HAVE **GROWING FINANCIAL, REPUTATIONAL AND LEGAL NEGATIVE CONSEQUENCES**



IOT CONTRIBUTION TO CYBER SECURITY RISK



MAIN RISKS OVERVIEW

SECURITY IS AN AFTERTHOUGHT

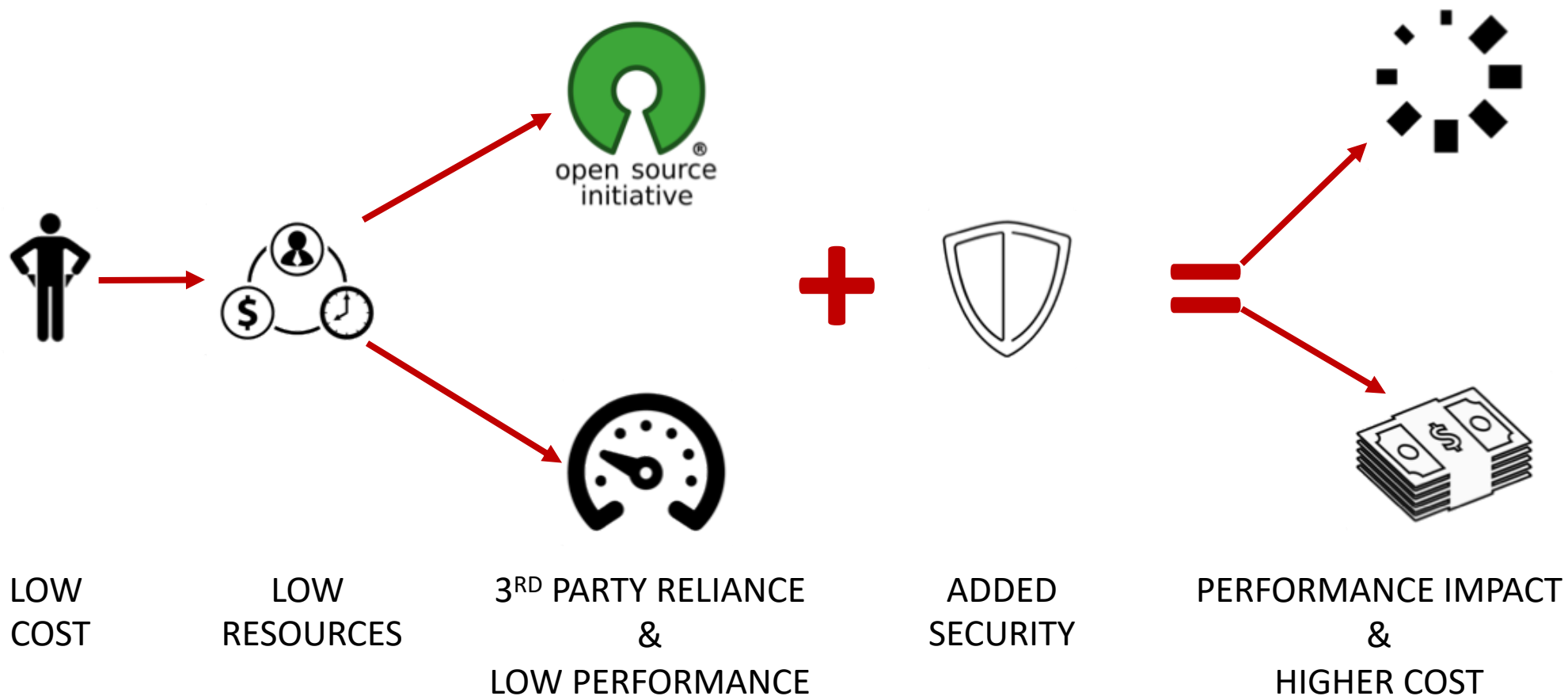
EMBEDDED VULNERABILITIES

EMBEDDED BACKDOORS

UNSUPPORTED DEVICES

UNPATCHABLE DEVICES

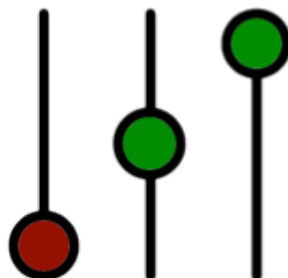
SECURITY IS AN AFTERTHOUGHT



EMBEDDED VULNERABILITIES



OLD LIBRARIES
& COMPONENTS



BAD CONFIGURATION



OPEN PORTS

EMBEDDED BACKDOORS

P A S S W O R D 1 2 3

DEFAULT PASSWORDS



DEBUG FUNCTIONS

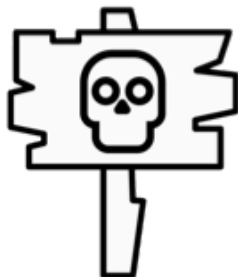


"VENDOR" ACCESS

UNSUPPORTED DEVICES



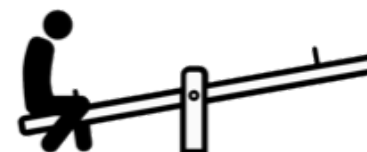
SHORT LIFESPAN



VENDOR
CLOSED

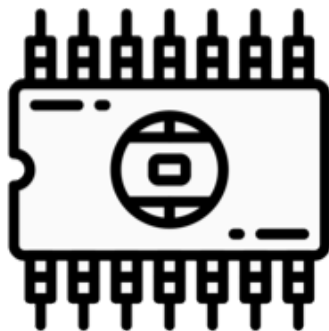


DIFFICULT
TO UPDATE



NO SUPPORT HELP

UNPATCHABLE DEVICES



ROM BASED
VULNERABILITY



OUT OF REACH



PHYSICAL DANGER

MAIN IMPACTS OVERVIEW

USED AS A BOT/DOS

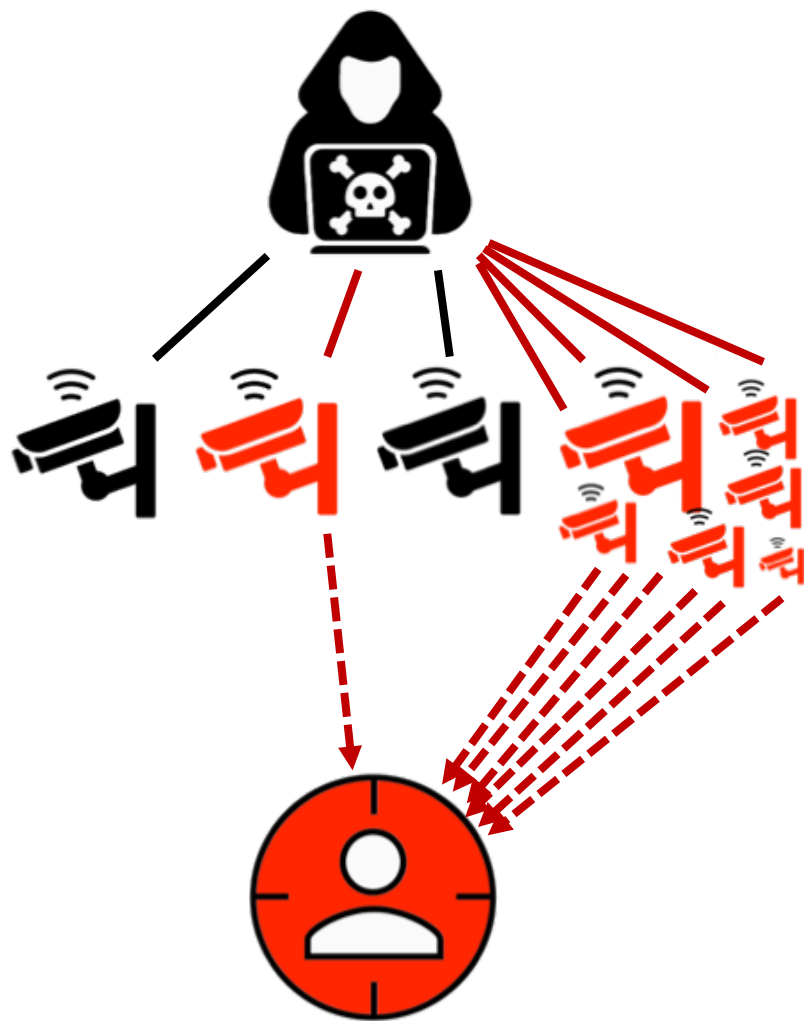
USED AS AN ENTRY TO YOUR NETWORK

USED TO SPY/ATTACK YOU

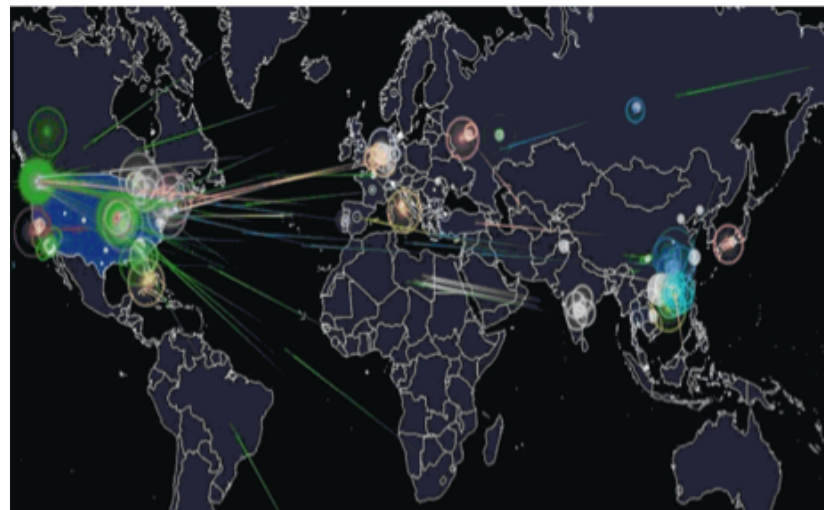
PHYSICAL IMPACT

PRODUCT DESTRUCTION

IOT USED AS A BOT



MIRAI, GAFGYT, AIDRA

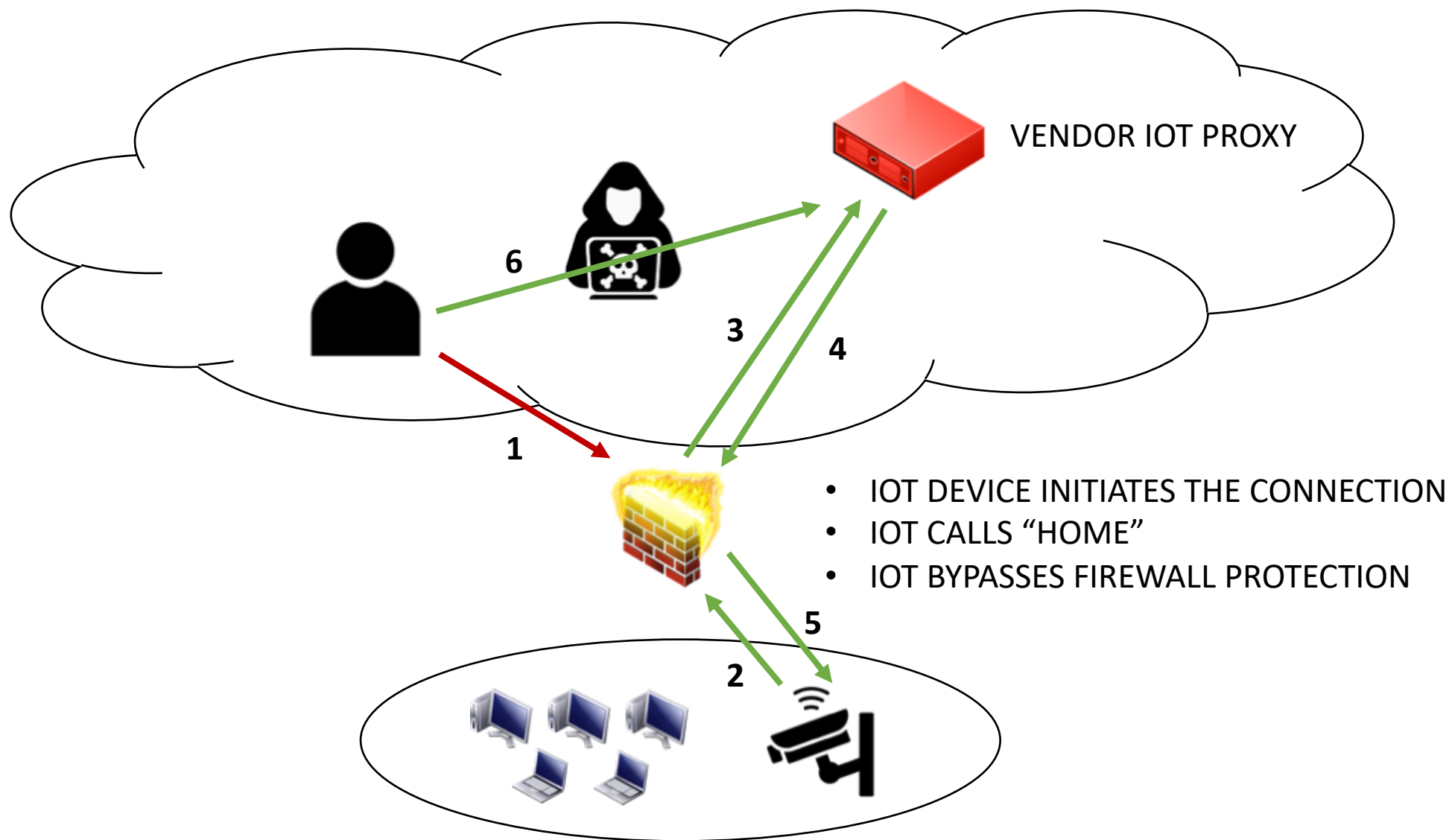


MIRAI

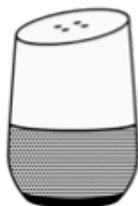
- TELNET OPEN
- 61 DEFAULT PASSWORDS
- 1TBPS

- ROUTERS, IP CAMERAS, ETC.

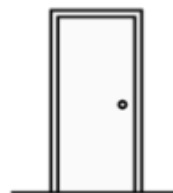
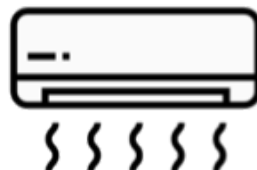
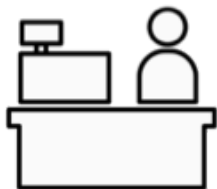
IOT USED TO ACCESS NETWORK



IOT USED TO SPY/ATTACK YOU



IOT PHYSICAL IMPACT



IOT SELF DESTRUCT BUTTON



MIKROTIK ROUTER
RUSSIAN GOOD SAMARITAN PATCH



NOKIA HEALTH
SCALE REFUNDED AND DISABLED

IOT SECURITY OVERVIEW

TRAFFIC ANALYSIS

CODE ANALYSIS

FIRMWARE ANALYSIS

ONLINE SEARCH / SHODAN

SECURITY DESIGN BEST PRACTISE

TRAFFIC ANALYSIS

- INTERCEPT TRAFFIC
- LOOK AT PASSWORDS SENT
- LOOK AT ENCRYPTION
- LOOK AT TOKENS



USER



VENDOR

CODE ANALYSIS

- CONNECT TO THE DEVICE
- INSPECT SOFTWARE INSTALLED
- LOOK AT SCRIPTS
- STRINGS IN BINARY



USER

- NO DEFAULT PASSWORD
- NO BACKDOOR FOR SUPPORT
- CODE ANALYSIS TOOLS
- THREAT ANALYSIS TOOLS



VENDOR

FIRMWARE ANALYSIS

- EXTRACT FIRMWARE
- DISASSEMBLE FIRMWARE
- REVIEW MAIN FUNCTIONS

- FIRMWARE ORIGIN?
- ANALYSE SOURCE CODE
- PENTEST RESULTS REVIEW



USER



VENDOR

ONLINE SEARCH

- GOOGLE SEARCH VENDOR
- SHODAN SEARCH DEVICE AND IP
- IS YOUR IP VULNERABLE?

- SHODAN SEARCH DEVICE
- WHITE PAPER REVIEW
- THREAT INTELLIGENCE



USER



VENDOR

UK CODE OF PRACTICE FOR CONSUMER IOT SECURITY



VENDOR

1. NO DEFAULT PASSWORDS
2. IMPLEMENT A VULNERABILITY DISCLOSURE POLICY
3. KEEP SOFTWARE UPDATED
4. SECURELY STORE CREDENTIALS AND SECURITY-SENSITIVE DATA
5. COMMUNICATE SECURELY
6. MINIMISE EXPOSED ATTACK SURFACES
7. ENSURE SOFTWARE INTEGRITY
8. ENSURE THAT PERSONAL DATA IS PROTECTED
9. MAKE SYSTEMS RESILIENT TO OUTAGES
10. MONITOR SYSTEM TELEMETRY DATA
11. MAKE IT EASY FOR CONSUMERS TO DELETE PERSONAL DATA
12. MAKE INSTALLATION AND MAINTENANCE OF DEVICES EASY
13. VALIDATE INPUT DATA

<https://www.gov.uk/government/publications/secure-by-design>

FUTURE OF IOT

IOT FRAMEWORK

IOT REGULATION

IOT BREACH FINES

IOT INTEGRATION

IOT FUSION

ABOUT ELYSIUMSECURITY LTD.

ELYSIUMSECURITY provides a portfolio of **Strategic** and **Tactical Services** to help companies **protect** and **respond** against **Cyber Security Threats**. We differentiate ourselves by offering **discreet, tailored** and **specialized** engagements.

ELYSIUMSECURITY provides **high level expertise** gathered through years of **best practices experience** in large international companies allowing us to provide advice best suited to your **business operational model** and **priorities**.

ELYSIUMSECURITY operates in **Mauritius** and in **Europe**, a ***boutique style*** approach means we can **easily adapt** to your business operational model and requirements to provide a **personalized service** that fits your working environment.

ELYSIUMSECURITY provides **practical expertise** to **identify vulnerabilities**, assess their **risks** and **impact**, **remediate** those risks, **prepare** and **respond** to incidents as well as raise **security awareness** through an organization.

